

HARSH SOLANKI

AI TRANSFORMATION & APPLIED AI LEADER

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SUMMARY

AI transformation and Applied AI leader with 10+ years delivering enterprise technology from use-case prioritization through governed production adoption. Lead a 7-engineer Applied AI team at Deriv Dubai across intelligent automation, retention, omnichannel engagement and GenAI enablement. Previously helped build Target's enterprise ML/data platform serving 1,500+ monthly data scientists and 100k+ workloads. Blend roadmap/program delivery, stakeholder adoption, responsible AI and hands-on architecture across GenAI, agentic systems, computer vision, distributed ML and cloud platforms.

10+ YEARS | 7-ENGINEER AI TEAM | 1,500+ MONTHLY USERS | 100K+ WORKLOADS | 3 US PATENTS (TEAM)

EXPERIENCE

Lead, Applied AI Engineering | Deriv (Regent Markets) | Dubai, UAE 07/2025 - Present

- **AI transformation & program leadership** - Lead a 7-engineer portfolio spanning retention/VIP, partner marketing, customer engagement and AI platform initiatives. Prioritize use cases with Product/business teams, translate Legal/Security requirements into implementation guardrails, and drive work from discovery/PoC through architecture review, rollout, adoption and operating metrics.
- **Praxis / Event360 - real-time intelligent automation** - Moved lifecycle evaluation from overnight batch to <5 minutes using BigQuery, NATS JetStream, Redis and Temporal durable workflows for retries, timers, signals and human approvals. Kept deterministic eligibility separate from model-generated content so AI assists decisions without becoming the authority for who is contacted.
- **EkaConnect / 361-Inboxes + CRM transformation** - Connected lifecycle decisions to one consent-aware engagement layer across WhatsApp, Telegram, Instagram, Messenger and email with canonical conversations, idempotent dispatch and explicit delivery/reconciliation states. Designed a phased HubSpot-to-owned-CRM target architecture around identity, journeys, lead/opportunity workflows, attribution and one-writer cutovers.
- **Cerebro - AI-assisted retention & customer experience** - Operate and integrate a FastAPI/Next.js/Supabase platform that rebuilds a ~1.2M-client BigQuery segmentation snapshot nightly and turns it into prioritized retention/VIP work queues, AI call preparation, outreach drafts, review intelligence, consent-aware voice readiness and management funnel reporting.
- **Postly - GenAI adoption from pilot to product** - Built a governed internal/partner content workflow combining trusted-source grounding, multimodal generation, deterministic rendering, versioned policy checks, exact-artifact approval and reconciled publishing. Hard compliance decisions remain outside prompts and elevated-risk actions stay human-gated.
- **Responsible AI, privacy, cybersecurity & provider governance** - Use Microsoft Presidio plus deterministic allowlists, redaction/tokenization before external LLM calls, sanitized LangFuse telemetry and human/policy gates for consequential writes. Hardened six AI products to zero outstanding critical vulnerabilities and standardized evaluation of Claude/GPT/Gemini, cloud AI, eventing and workflow options against security, cost, latency, quality and lock-in.

Senior -> Lead ML Engineer | Target | Bengaluru, India 09/2021 - 07/2025

- **Enterprise ML platform adoption** - Helped build Target's self-service ML/data platform across Kernels-as-a-Service, Feature Store, MLflow lifecycle/GitOps and DEMS event telemetry, supporting 1,500+ monthly active data scientists and 100k+ lifetime Spark/training workloads. Drove adoption through SDKs, standard contracts and declarative configuration rather than one-off platform tickets.
- **Kernels-as-a-Service (KaaS)** - Part of the team behind patent US 2023/0229438 A1; enabled one Jupyter experience across Vertex AI, BigRed3 YARN, TAP Kubernetes, Ray and GPUs. Owned BigRed3/Argus ProcessProxy integration, developer experience, token propagation and long-lived authentication refresh across the notebook path.
- **Feature Store & measurable ML quality** - Built API/SDK and online-serving components on Trino + Iceberg with Redis/ScyllaDB stores and feature lineage into MLflow; train-serve skew dropped 3.1% -> 0.4% and PII-feature impact analysis became answerable in minutes.
- **MLflow / GitOps / DEMS governance** - Contributed SSO, model-card enforcement and event-driven canary/rollback; co-designed targetml.yml to bind models, features, deployment and monitoring, and plumbed DEMS/Lasso telemetry for drift, SLA analytics and lineage.

- **Nexus + Agentic AI / LLMops** - Shipped permission-aware RAG/agent workflows using R2R, LangGraph, LangChain, Guardrails and LangFuse; standardized Claude/GPT/Gemini routing, tool contracts, structured outputs, prompt/model versioning, evals and retries/fallbacks. Added Presidio/context allowlists before external calls and evaluated fine-tuning vs RAG plus quantization trade-offs for open-source model serving. Hired and grew an 8-engineer team during the GenAI lead phase.

Machine Learning Engineer | Quantiphi | Bengaluru, India

09/2020 - 09/2021

- **Retail loss prevention & organized retail crime** - Developed computer-vision systems addressing theft, fraud and organized retail crime across store environments, combining video, POS and transaction signals to surface suspicious behavior for review.
- **Video understanding & miss-scan detection** - Built human-activity/video-analysis pipelines and self-checkout miss-scan models to detect non-compliant actions at POS; explored multimodal/generative techniques for richer behavioral interpretation while keeping final decisions grounded in observable evidence.
- **Person re-identification & distributed AI** - Designed cross-camera person re-identification/tracking approaches for movement across store zones and locations, balancing recognition quality, latency and deployment constraints for distributed/edge inference.
- **Gift-card tampering & multimodal fraud** - Developed detection approaches combining visual cues with transaction/context metadata to identify gift-card tampering and suspicious activation patterns rather than relying on a single signal.
- **MLOps, feature-store foundations & recovery** - Delivered forecasting and computer-vision systems on GCP/AWS and contributed to in-house feature-storage/model workflow foundations for scalable training and serving. Led a production silent-failure post-mortem after upstream schema changes caused 11 days of drift, turning the lessons into stronger data contracts, monitoring and recovery patterns.

Business Analyst | Amazon | India

03/2015 - 09/2020

- **Business transformation foundation** - Spent 5+ years in Retail and Operations analytics building decision-support dashboards and experimentation frameworks, developing the ability to translate operational problems into measurable technology interventions.

LEADERSHIP & TRANSFORMATION

- **Stakeholder engagement & change adoption** - Run architecture/design reviews, translate business, Product, Legal and Security requirements into implementation guardrails, and create reusable patterns that help teams adopt AI safely rather than depend on one-off expert support.
- **Innovation, pilots & technology evaluation** - Take emerging AI ideas through problem framing, PoC, architecture review and production readiness; compare build/buy and provider options on security, quality, cost, latency, operational complexity and lock-in before broader adoption.
- **Engineering leadership & communication** - Lead and mentor engineering teams, communicate technical trade-offs to non-specialist stakeholders, and actively write/speak on production MLOps, feature stores, GenAI and platform engineering.

SKILLS

AI Transformation: roadmaps, use-case prioritization, PoCs/pilots, business cases, cross-functional delivery, rollout governance, change adoption

GenAI / Agentic AI: RAG, LangChain, LangGraph, R2R, MCP, tool/function calling, Transformers, LLM fine-tuning & quantization, evals

Computer Vision: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, person re-identification, video analytics, multimodal AI, OpenVINO

Distributed AI / MLOps: MLflow, Kubeflow/KFP, Ray, KServe-style serving, Temporal, NATS JetStream, Kafka, WebSockets, canary/rollback

Data / Cloud: BigQuery, Spark, Trino/Iceberg, Redis, ScyllaDB, PostgreSQL, GCP/Vertex/GKE, AWS/SageMaker/ECS, Azure/OpenAI, Docker/Kubernetes, Helm/Terraform

Responsible AI / Security: Microsoft Presidio, PII redaction/tokenization, prompt/model versioning, LangFuse, human/policy gates, OAuth2/OIDC/JWT, Vault/Key Vault, tenant isolation

CREDENTIALS, PATENTS & EDUCATION

Certifications: AWS Certified Machine Learning - Specialty | AWS Certified Solutions Architect - Associate | AI Fluency Framework & Foundations

Patents (team): KaaS (US 2023/0229438 A1) | DEMS/Lasso (US 2021/0124786 A1) | Ad-hoc Data Exploration Tool (US 11,675,814)

Education: PGP Data Science & ML - U. of Chicago Graham School | B.E. - Rajiv Gandhi University

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